

Automated Conveyor Sorting System

Automation System Design – Design and Operation Report

Prepared by: B.Gaganalakshmi | Submitted for: Internship Task

1. Introduction

This report presents the design of an automated conveyor sorting system for a light manufacturing or packaging line. The objective is to transport products along a belt, automatically inspect each item against a defined quality criterion, and sort items into accepted and rejected streams with minimal human intervention. The design covers the system architecture, the sensors and actuators used, the control logic executed by the controller, and the safety features that protect operators and equipment. A Programmable Logic Controller (PLC) is used as the controller because it is the industry-standard choice for reliable, real-time machine control; a microcontroller such as an Arduino or ESP32 could be substituted for a small-scale or prototype build.

2. System Overview

The system is built from three interacting subsystems: a sensing subsystem that gathers information about the items and the machine state, a control subsystem (the PLC) that makes decisions, and an actuation subsystem that moves the belt and diverts items. In normal operation, an item placed on the belt is detected by an entry sensor, which increments an item counter. The item is then conveyed to an inspection point, where a sensor checks whether it meets the pass criterion – for example, the correct height, or an unbroken through-beam. Items that pass continue to the output end of the line; items that fail are pushed by a pneumatic diverter into a reject bin. A three-colour stack light and buzzer communicate the machine status to the operator. Figure 1 shows how the sensors and actuators connect to the PLC.

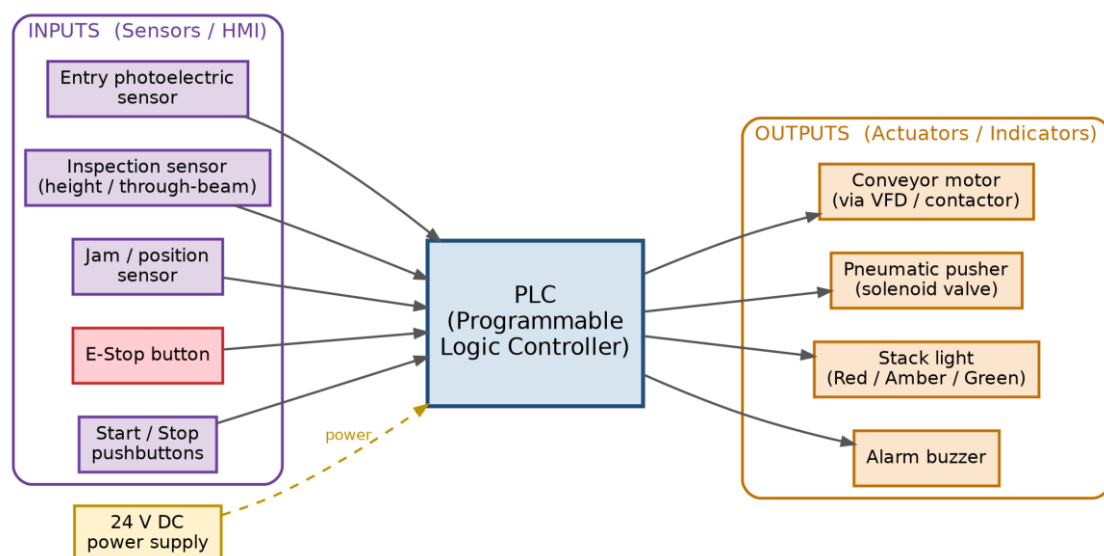


Figure 1 – System architecture. Inputs (sensors and operator controls) feed the PLC, which drives the actuators and indicators.

3. Hardware Components

The main hardware is summarised in Table 1. Component types are given as representative examples; exact models would be chosen during detailed design to match the product size, belt speed and operating environment.

Component	Type / example	Function
Conveyor motor	Geared 3-phase AC motor with VFD (or 24 V DC gear motor)	Drives the belt; the VFD allows belt speed to be adjusted.
Entry sensor	Diffuse photoelectric sensor	Detects an item entering the line and triggers the item counter.
Inspection sensor	Through-beam photoelectric or laser height sensor	Checks each item against the pass/fail criterion.
Jam / position sensor	Retro-reflective photoelectric sensor + timer	Confirms items clear the line and flags a jam on timeout.
Controller	Programmable Logic Controller (PLC)	Runs the control program: reads inputs and drives outputs in real time.
Diverter	Pneumatic cylinder with solenoid valve	Pushes failed items into the reject bin.
Indicators	Three-colour stack light and buzzer	Show machine status (green / amber / red) and raise alarms.
Operator controls	Start / Stop pushbuttons and E-stop	Allow manual control and emergency shutdown.
Power supply	24 V DC PSU	Powers the sensors, PLC I/O and indicators.

Table 1 – Principal components and their functions.

4. Control Logic and Sequence of Operation

The control program runs as a continuous loop, summarised in the flowchart in Figure 2 (Appendix A). On power-up the PLC initialises its inputs, outputs and counters, then checks the safety circuit: the emergency-stop must be released and no fault present. If the safety check fails, the controller shows a red status light and waits for the operator to clear the fault and reset. Once the system is safe and the operator presses START, the conveyor motor runs and the status light turns green.

In the main loop the PLC monitors the entry sensor. When an item is detected, the item counter is incremented and the item is conveyed to the inspection sensor. The controller then evaluates the pass/fail criterion. A failing item triggers the pneumatic pusher, which diverts it into the reject bin and increments a separate reject counter; a passing item continues to the output and increments the pass counter. After each item the controller checks for a jam, fault or emergency-stop. If none is present, the loop repeats; if a fault is detected, the motor stops, the buzzer sounds and the status light turns red, bringing the machine to a safe halt. Jam detection is implemented with a timer – if an item does not clear a sensor within an expected time window, the controller treats it as a jam.

5. Safety Features

- A hardwired emergency-stop cuts motor power independently of the program logic, so the belt stops even if the controller is unresponsive.
- Motor overload and thermal protection is provided through the contactor or VFD to prevent damage from a stalled belt.
- Jam detection stops the line and raises an alarm if an item fails to clear a sensor within the expected time.
- Outputs are fail-safe: on power loss or fault the motor stops and the pneumatic pusher returns to its home position.
- Fixed guarding (and an optional light curtain) protects operators from moving parts, while the stack light gives a clear visual indication of machine state.

6. Possible Enhancements

- Replace the single inspection sensor with a machine-vision camera to check labels, dimensions or defects.
- Add data logging and an IoT or SCADA dashboard to track throughput, reject rate and uptime.
- Vary belt speed automatically with demand to balance the line and save energy.
- Add further diverters to sort items into more than two categories.
- Monitor motor current and vibration to enable predictive maintenance.

7. Conclusion

The proposed design meets the goal of an automated conveyor system that transports, inspects and sorts items reliably and safely using standard industrial components. The control logic is straightforward to implement on a PLC and is built around clear decision points and a fail-safe stop condition, while the modular hardware makes the system easy to extend – for example, by adding machine vision or data logging – as requirements grow.

Appendix A – Control-Logic Flowchart

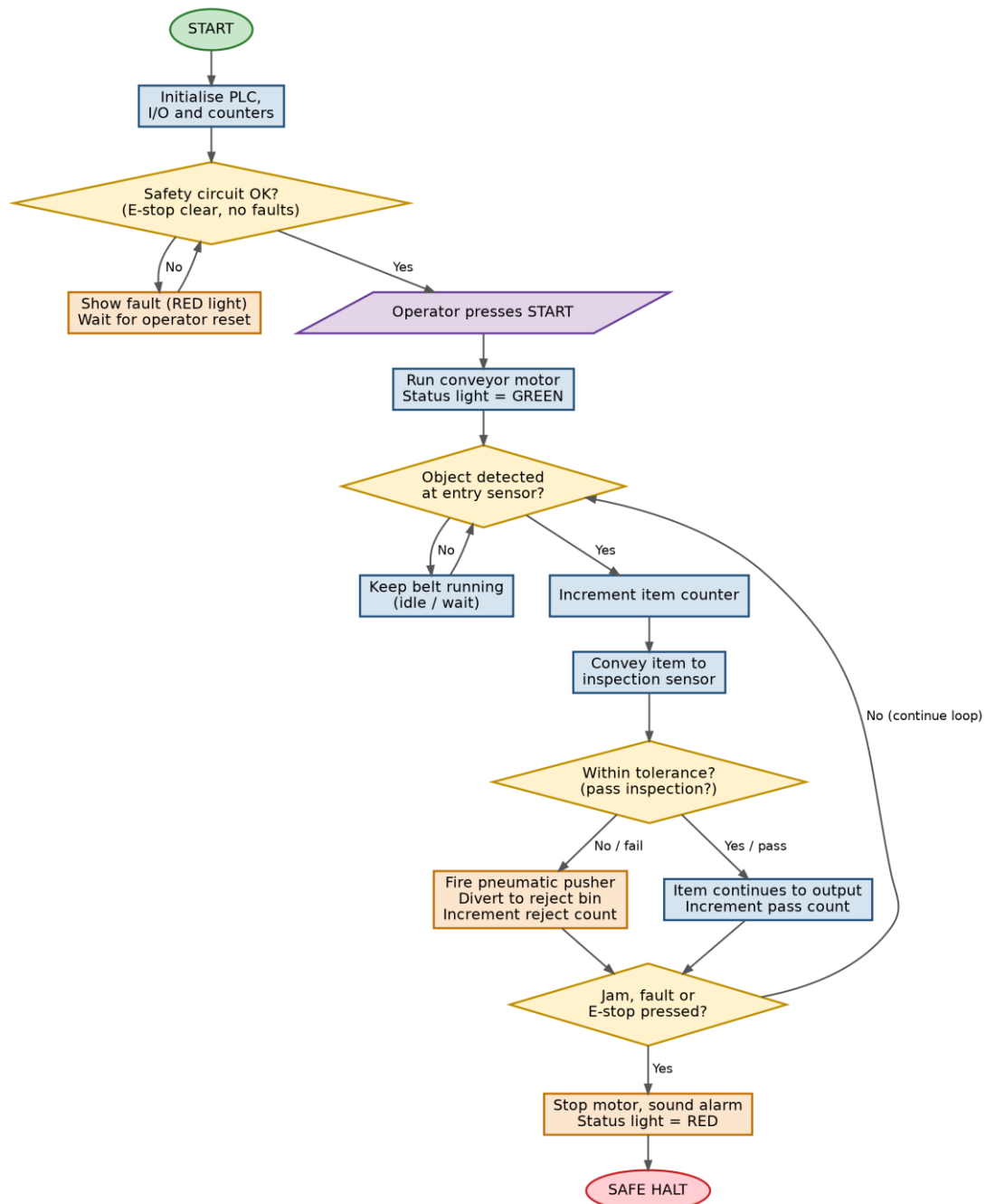


Figure 2 – Control-logic flowchart for the conveyor sorting sequence.